

Alpha4 Reconfigurable Design

Goal

Using extra input control signals (OS and simd) to easily reconfigure the systolic array to perform computation using Weight Stationary mode or Output Stationary mode, and SIMD (2 weights computed in 1 PE).

How to Run

The following simulations all use the same designs of *mac*, *mac_tile*, *mac_row* and *mac_array*. Computation modes are changed by delivering different combinations of signals to *OS* and *simd* ports of *mac_array*.

1. 4-bit activation, 4-bit weight, Weight Stationary

Compile *.v files under *./verilog/WS* directory, run simulation with *core_tb_4bit.v* and data in *./datafiles/4bit_data*.

2. 2-bit activation, 4-bit weight, Weight Stationary, SIMD

Still under *./verilog/WS* directory, run simulation with *core_tb.v* and data in *./datafiles/2bit_data*.

3. 4-bit activation, 4-bit weight Output Stationary

Compile *.v files under *./verilog/OS* directory, run simulation with *./datafiles/os_4bit_data*. We observed that we got the first four rows of output match *./datafiles/os_4bit_data/out.txt*. We believe data error is the cause.

4. 2-bit activation, 4-bit weight, Output Stationary, SIMD

Testbench in progress. Logic design has been done and ports (specifically, *in_n*) have been reserved.